AutoXRD: Autonomous LLM Agents and Comprehensive Evaluation for Powder Diffraction Analysis

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ABSTRACT

Powder X-ray diffraction (XRD) is central to materials characterization, yet reliable end-to-end automation remains challenging. An XRD agent must interpret diffraction evidence, operate refinement software, manage coupled parameters in a defensible order, and distinguish numerical improvement from physical validity. In this paper, we propose AutoXRD, an autonomous large language model (LLM) agent framework that organizes powder-XRD analysis as stepwise refinement, grounds actions in observed evidence, and applies deterministic crystallographic and physical checks before accepting results. We further introduce XRDBench with two complementary tracks. XRDBench-QA contains 100 bounded diagnostic tasks that isolate scientific reasoning and decision-making, whereas XRDBench-E2E contains 34 executable workflows that test whether agents can compose these capabilities into complete analyses requiring file inspection, crystallographic-software execution, iterative refinement, evidence preservation, and reporting. We evaluate ten recent LLMs across 1,340 model–task runs. Models average only 57.8 out of 100, falling from 61.9 on XRDBench-QA to 53.7 on XRDBench-E2E. They perform best on refinement-history assessment and result acceptance, but remain substantially weaker on refinement-action selection, phase quantification, indexing, and Rietveld refinement. GPT-5.6 Sol achieves the highest overall score of 81.1, GPT-5.6 Terra the highest E2E point estimate of 81.0, and GPT-5.6 Luna the best score–cost trade-off. Ablations show that all six AutoXRD components consistently improve performance, supporting the framework design. Finally, execution-trace analysis reveals recurring failures in coupled-parameter control, quantitative reasoning, evidence preservation, and workflow termination, motivating stronger scientific constraints, uncertainty-aware decisions, and more efficient planning.

stephen-smj.github.io/autoxrd_page

CCS CONCEPTS

• **Computing methodologies** → **Intelligent agents**; *Natural language processing*; • **Applied computing** → *Physical sciences and engineering*.

KEYWORDS

scientific agents, tool use, powder diffraction, Rietveld refinement, agent evaluation, AI for science

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1 INTRODUCTION

Powder X-ray diffraction (XRD) compresses three-dimensional reciprocal-space information into a one-dimensional intensity profile. Peak overlap, background, instrumental aberrations, preferred orientation, finite crystallite size, strain, and phase mixtures can therefore produce similar observations. Rietveld refinement addresses overlap by fitting the complete profile [25, 41], but it does not turn analysis into an unconstrained curve-fitting problem. The parameters governing the crystal structure, specimen, and instrument are strongly coupled; successful practice depends on releasing them in a defensible order and inspecting the observed, calculated, and difference patterns after each stage [34, 35].

These requirements expose two related gaps in current language-agent research. First, tool-using language models can interleave reasoning and external actions [29, 39], and domain agents have automated parts of chemistry and experimentation [1, 2], but a general tool loop provides no scientific guarantee. A syntactically valid action may jointly release non-identifiable parameters, a lower R_{wp} may arise from an incorrect structural or peak-profile hypothesis, and a fluent report may conceal an impossible occupancy or an unresolved phase. Second, current XRD-agent evaluation is not yet systematic. Existing studies demonstrate the feasibility of LLM-assisted Rietveld refinement [18, 30], but evaluations are often based on selected cases, isolated questions, or final fit statistics. No single criterion is sufficient: exact matching cannot accommodate valid open-ended analyses, fit metrics do not establish physical correctness, and judging only the final report cannot reveal whether a failure arose from scientific reasoning, software execution, missing artifacts, or poor termination. The field therefore lacks a unified protocol that tests both individual XRD decisions and complete executable workflows while measuring scientific quality, execution validity, efficiency, and failure behavior across models.

Our abstraction treats refinement as a stepwise graph of scientific decisions. A node stores a complete fit state; an edge stores one structured action, the evidence that supports it, and predictions that can falsify its rationale. Numerical software computes the candidate child state, but deterministic checks—not the language model—decide whether the step is accepted. Rejected edges remain first-class evidence, enabling failure analysis and future policy distillation.

Figure 1 summarizes AutoXRD. At runtime, the language model maintains session context and memory, plans with ten XRD-specific skills, and invokes permissioned tools within a sandbox. The skills translate domain procedures into executable scientific plans that specify the refinement stage, parameters or code, numerical bounds, required evidence, and expected outcomes. The scientific action layer parses diffraction patterns and structures, enforces parameter constraints, executes FullProf, GSAS-II, or task-appropriate numerical routines, and stores the resulting artifacts. Pattern-quality,

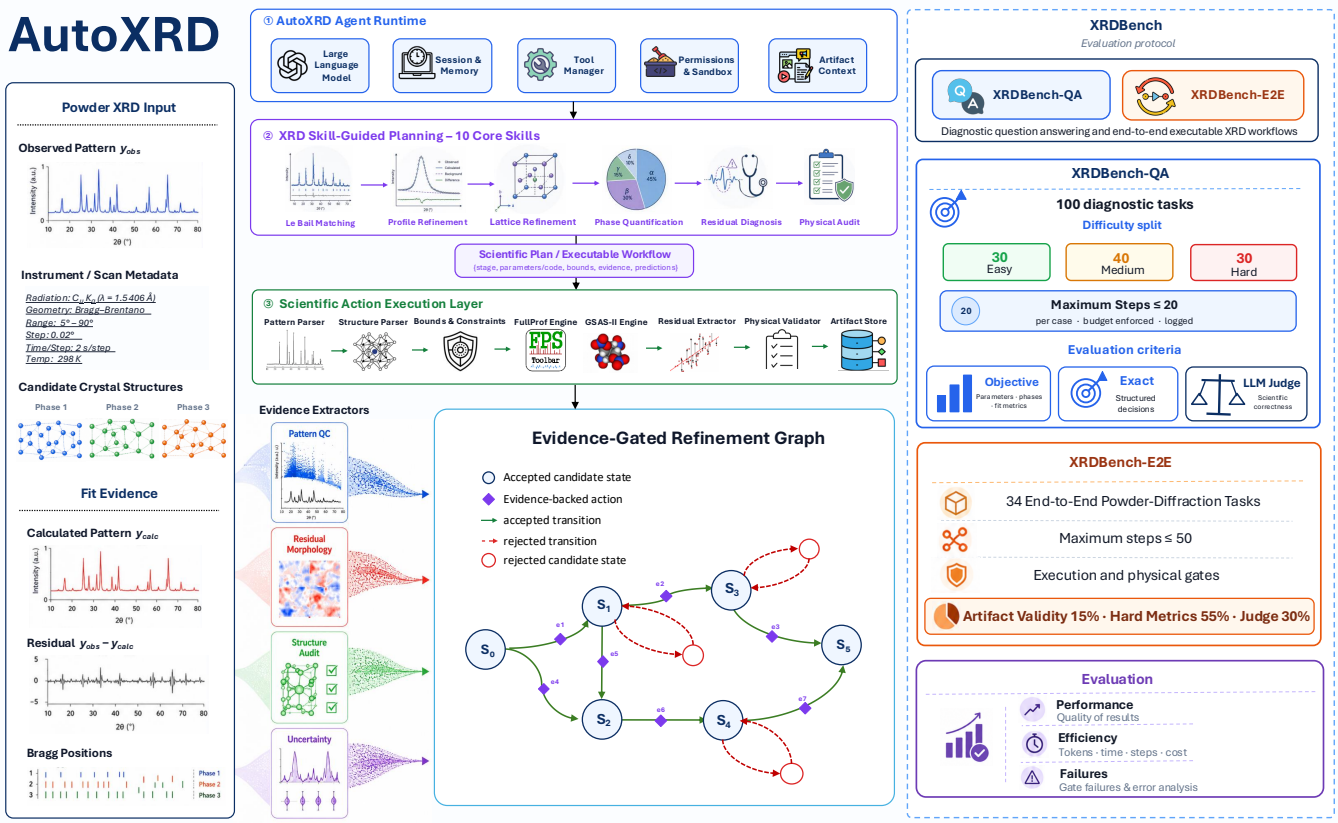


Figure 1: Overview of AutoXRD and the XRDBench evaluation protocol. AutoXRD combines an autonomous agent runtime, skill-guided powder-diffraction workflows, executable crystallographic backends, and evidence-based scientific validation. XRDBench evaluates diagnostic reasoning through XRDBench-QA and autonomous end-to-end XRD analysis through XRDBench-E2E.

residual, structural, and uncertainty evidence is then used to accept or reject candidate transitions in an evidence-gated refinement graph.

The right panel of Figure 1 presents XRDBench, designed to address the evaluation gap through two complementary tracks. XRDBench-QA contains 100 bounded diagnostic tasks that isolate scientific reasoning and decisions, enabling reliable exact, objective, and reference-guided assessment. XRDBench-E2E contains 34 executable workflows over measured patterns, requiring agents to inspect inputs, implement and run numerical analyses, validate intermediate results, and deliver auditable artifacts. Neither track is sufficient alone: the former diagnoses what an agent understands, whereas the latter tests whether those capabilities can be composed into a functioning analysis. Together they evaluate the same XRD process at different degrees of integration and report performance, efficiency, and failure behavior under a common protocol. Using XRDBench, we evaluate ten recent LLMs in 1,340 agent runs. Average performance decreases from 61.9 on XRDBench-QA to 53.7 on XRDBench-E2E, and the two track scores are only moderately correlated ($r = 0.56$, $\rho = 0.65$), showing that diagnostic competence does not reliably translate into autonomous execution. Models perform best on refinement-history assessment and result acceptance, but remain substantially weaker on refinement-action selection, phase quantification, indexing, and Rietveld refinement.

To summarize, our main contributions are:

- **A scientifically grounded autonomous XRD agent.** We introduce AutoXRD, which combines XRD skill-guided planning, executable crystallographic backends, and deterministic scientific validation. Rather than accepting a step merely because the software runs or the fit residual decreases, AutoXRD requires the predicted evidence change and crystallographic checks to pass, while preserving both accepted and rejected actions in an auditable refinement trajectory.
- **A two-track benchmark for XRD reasoning and execution.** We introduce XRDBench, comprising 100 bounded XRDBench-QA diagnostic tasks and 34 executable XRDBench-E2E workflows across eleven powder-diffraction task families. Its complementary evaluation protocols measure scientific decisions, numerical results, physical validity, artifact completeness, efficiency, and failure behavior.
- **A large-scale empirical study of LLMs for XRD.** We evaluate ten recent LLMs across 1,340 agent runs, 22,135 model turns, and 625.6M selected-trajectory tokens. Our analysis shows that

diagnostic performance does not reliably transfer to autonomous execution and identifies refinement-action selection, phase quantification, indexing, and Rietveld refinement as major remaining bottlenecks.

2 BACKGROUND AND RELATED WORK

2.1 Rietveld Refinement and Its Failure Modes

Let x_i denote the diffraction coordinate (e.g., 2θ, time of flight, or d-spacing), y_i the observed intensity, σ_i its standard uncertainty, and $\hat{y}_i(\theta)$ the profile calculated from parameters θ. Classical Rietveld refinement minimizes weighted least squares [25]:

$$\theta^* = \arg\min_{\theta\in\Theta} \chi^2(\theta), \qquad \chi^2(\theta) = \sum_{i=1}^{n} w_i \big[y_i - \hat{y}_i(\theta)\big]^2, \tag{1}$$

where $w_i = \sigma_i^{-2}$ when uncertainties are available. The vector θ includes scale, background, zero shift, lattice, profile, atomic coordinates, displacement factors, occupancies, preferred orientation, and microstructure terms. Le Bail profile matching can align the cell, background, and peak profile before atomic parameters are exposed [15]. For a fully crystalline mixture with all relevant phases included in the refinement model, phase weight fractions can be derived from refined scale factors and phase-specific ZMV terms, where Z, M, and V denote formula units per cell, formula mass, and unit-cell volume [12]. Preferred orientation and line shape require specialized models [8, 32].

No scalar fit index establishes structural correctness. For example, the weighted-profile factor

$$R_{\mathrm{wp}} = 100\sqrt{\frac{\sum_i w_i(y_i - \hat{y}_i)^2}{\sum_i w_i y_i^2}} \tag{2}$$

depends on data quality and uncertainty conventions; a wrong model on weak data may achieve a lower value than an excellent model on precise data [34]. This motivates explicit checks of residual structure, parameter correlations, chemistry, bounds, and unresolved peaks.

2.2 Automation of XRD Analysis

Crystallographic engines such as FullProf [27] and GSAS-II [36] provide mature numerical capabilities. Automation has progressed from high-throughput wrappers (SrRietveld) [33], to expert systems (AutoFP) [7], reinforcement learning (PowderBot) [10], black-box optimization (BBO-Rietveld) [24], derivative-guided recipe selection [35], and learned parameter prediction [23]. Robotic PXRD systems have integrated sample preparation, measurement, and automated analysis [40].

Phase identification is related but distinct. Deep models can learn multi-phase labels from synthetic patterns [16, 17]; CrystalShift uses symmetry-constrained pseudo-refinement and Bayesian comparison [5]; Dara searches and refines multiple phase combinations [9]; and RADAR-PD couples mismatch-tolerant prediction with Rietveld verification [31]. Recent end-to-end pipelines broaden this scope: AGAPI-XRD combines database matching, a fine-tuned structure generator, and GSAS-II/BGMN refinement [3], while MatDiffract couples hierarchical retrieval with full-pattern refinement and quantitative phase fitting [37]. COD [11], opXRD [13], SimXRD-4M [4], and SIMPOD [26] improve data availability. These advances automate engines, prediction, or search. Our focus is complementary: constraining and evaluating the sequence of decisions made by a language agent operating scientific software.

Table 1 compares observable functionality rather than project-specific terminology. Prior systems cover important but separated parts of the problem: SrRietveld, AutoFP, PowderBot, and BBO-Rietveld automate refinement; CrystalShift and Dara address phase ambiguity; and Rongzai Agent and AgentBuild introduce LLM-based control. Rongzai compares four base models on five neutron-diffraction samples and records iterative refinement, while AgentBuild evaluates reports and trajectories with an expert-authored rubric and curriculum. AGAPI-XRD and MatDiffract benchmark automated structure identification and refinement over hundreds of patterns. None provides a reusable benchmark spanning bounded scientific diagnosis and heterogeneous executable workflows with cross-model analysis of decisions, tool trajectories, and execution behavior. In contrast, XRDBench evaluates one process through two complementary parts: 100 QA tasks isolate scientific decisions and outcomes, while 34 state-changing workflows test their composition into executable analyses. Together they measure reasoning, scientific outcomes, execution evidence, physical validity, artifacts, trajectories, token use, wall time, and tool efficiency under a shared ten-model comparison.

2.3 Tool-Using and Scientific Agents

ReAct interleaves language reasoning with actions [39], while Toolformer learns when and how to call APIs [29]. AgentBench [19], τ-bench [38], and MLAgentBench [14] show that endpoint success alone can hide long-horizon failures and inconsistency. Scientific-agent work includes tool-augmented reasoning [21], chemistry agents [2], autonomous laboratory orchestration [1], and broad research automation [20]. ScienceAgentBench argues for authentic, task-level assessment before claims of end-to-end discovery [6].

AutoXRD applies this lesson to a narrow scientific domain with unusually sharp validity constraints. It differs from generic agents by making action schemas, falsifiable predictions, stage order, coupled-parameter exclusions, mandatory physical checks, and rejected steps part of the executable protocol.

3 PROBLEM FORMULATION

3.1 Refinement States and Structured Actions

An input instance is

$$\mathcal{I} = (\mathcal{D}, \mathcal{M}, C), \quad \mathcal{D} = \{(x_i, y_i, \sigma_i)\}_{i=1}^{n}, \tag{3}$$

where $\mathcal{M}$ is instrument metadata and C is a set of candidate structures. At step t, the agent observes

$$s_t = (\theta_t, q_t, r_t, v_t, c_t, O_t), \tag{4}$$

where $q_t = (R_{\mathrm{wp},t}, R_{\mathrm{exp},t}, \mathrm{GoF}_t)$ contains fit scalars, r_t contains residual features, v_t is a set of physical violations, c_t describes complexity and correlations, and O_t indexes backend output files. Here R_{wp} is the weighted-profile residual, R_{exp} is the statistically expected residual, and $\mathrm{GoF} = R_{\mathrm{wp}}/R_{\mathrm{exp}}$ denotes goodness of fit.

Table 1: Functional comparison with representative automated powder-diffraction systems. ✓ denotes a capability explicitly provided or evaluated in the cited work; ✗ means that the work does not report it. Residual-guided iteration uses localized or structured residual evidence; physical-validity checks go beyond a scalar fit index or optimizer bound. Public suites require reusable tasks and scoring protocols; complete telemetry reports tokens, wall time, and tool- or step-level measurements.

	Scientific analysis capabilities				Agent and evaluation capabilities					
System	Automated Rietveld refinement	Residual-guided iteration	Physical-validity checks	Multiple-hypothesis analysis	LLM tool agent	Auditable action trajectory	Public diagnostic suite	Public executable workflow suite	Cross-model evaluation	Complete agent telemetry
SrRietveld [33]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
AutoFP [7]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
PowderBot [10]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
BBO-Rietveld [24]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗
CrystalShift [5]	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗
Dara [9]	✓	✓	✗	✓	✗	✗	✗	✗	✗	✗
AGAPI-XRD [3]	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗
MatDiffract [37]	✓	✗	✗	✓	✗	✗	✗	✗	✗	✗
Rongzai Agent [18]	✓	✓	✓	✗	✓	✓	✗	✗	✓	✗
AgentBuild [30]	✓	✓	✓	✗	✓	✓	✗	✗	✗	✗
AutoXRD (ours)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

AutoXRD supports two complementary execution modes. In the validated refinement path, the agent proposes a structured action

$$a_t = (k_t, g_t, P_t, B_t, \mathcal{Z}_t, \Pi_t), \tag{5}$$

where k_t is an action kind, g_t a refinement stage, P_t a parameter subset, B_t optional bounds, $\mathcal{Z}_t$ cited evidence, and Π_t falsifiable predictions. Each evidence item is (f, v, τ, src): feature name, observed value, decision threshold, and data source. A prediction $\pi = (m, d, \delta)$ specifies metric m, required direction $d \in \{-1, +1\}$, and minimum effect $\delta \geq 0$. In the native code-agent path, actions may instead read, create, edit, or execute task files. These broader actions are retained verbatim in the trajectory and are evaluated through execution, artifact, and physical gates.

The deterministic backend produces a candidate

$$(s'_{t+1}, \mathcal{O}_{t+1}) = F(s_t, a_t; \mathcal{D}), \tag{6}$$

but the transition is not automatically accepted. This distinguishes three questions: (i) did the optimizer find a numerically better point; (ii) did the intervention change the feature predicted by its stated mechanism; and (iii) is the child state physically and statistically admissible?

3.2 Refinement Decision Graph

A refinement decision graph is $G = (V, \mathcal{E}^+, \mathcal{E}^-)$. Each $s \in V$ is a fit state, $\mathcal{E}^+$ contains accepted actions, and $\mathcal{E}^-$ contains rejected actions. Let $H(s, a, s') \in \{0, 1\}$ denote hard admissibility and

$$M(s, a, s') = \prod_{\pi \in \Pi(a)} \mathbb{I}\big[d_\pi\big(m_\pi(s') - m_\pi(s)\big) \geq \delta_\pi\big] \tag{7}$$

denote support for the action's expected outcome. The candidate child is accepted only if

$$\text{Accept}(s, a, s') = H(s, a, s')\, M(s, a, s') \mathbb{I}\big[U(s') - U(s) > 0\big] = 1. \tag{8}$$

Crucially, utility cannot compensate for $H = 0$ or $M = 0$.

4 AUTOXRD

4.1 Layered Agent Architecture

As shown in Figure 1, AutoXRD separates five responsibilities. The agent controller manages the model, tools, sessions, and budgets. An XRD procedure library provides ten skills: pattern quality control, structure validity checking, executable GSAS-II workflows, Le Bail initialization, validated PCR generation, staged refinement, residual extraction, residual diagnosis, result acceptance, and final scientific review. The XRD execution layer parses patterns and structures, invokes GSAS-II, FullProf, or task-appropriate numerical crystallography code, and extracts reproducible measurements. Pre-execution and post-execution checks enforce parameter, fit, residual, and physical constraints. Finally, a run-history store and report generator expose the complete sequence of accepted and rejected steps.

The language model is therefore a planner, implementer, and explainer, not the numerical optimizer or final scientific authority. This division follows a capability principle: use language models for context-dependent hypothesis selection, mature crystallographic engines for nonlinear optimization, and deterministic code for invariants that can be expressed exactly.

4.2 Deterministic Residual Evidence

For residuals $e_i = y_i - \hat{y}_i$, the feature extractor estimates robust scale

$$\hat{\sigma}_e = \frac{\text{median}_i\, |e_i - \text{median}(e)|}{0.67448975} \tag{9}$$

and standardized residuals $z_i = (e_i - \text{median}(e))/\hat{\sigma}_e$. Let $\tilde{e}_i = e_i - \bar{e}$, where $\bar{e} = n^{-1} \sum_i e_i$. The extractor reports normalized absolute residual, lag-one autocorrelation

$$\rho_1 = \frac{\sum_{i=1}^{n-1} \tilde{e}_i \tilde{e}_{i+1}}{\sum_{i=1}^{n} \tilde{e}_i^2}, \tag{10}$$

low/high-angle signed bias, the fraction of points in contiguous regions with $|z_i| \geq 5$, and an unexplained-peak ratio. These features make diagnoses testable. For example, a nearly angle-independent peak-position offset can motivate zero-shift refinement. Geometry-dependent specimen-displacement and calibration effects must first

be excluded, and the action must predict a minimum decrease in a position-bias feature. If R_{wp} decreases but the predicted feature does not, Equation 7 rejects the causal explanation.

4.3 Validated PCR Generation

FullProf PCR files are positional and contain codewords that can couple multiple parameters. The PCR validator and generator parses comment-anchored semantic slots into a catalog, freezes catalogued codeword groups, and releases only selectors permitted by k_t. It validates stage membership, selector coverage, constraints, phase count, pattern count, and mode before writing a new file. The source is never overwritten.

Let $\Gamma(k)$ be the allowed semantic selector set for action kind k and $\mathrm{cl}(P)$ the closure of P under existing constraint groups. Compilation is permitted only when

$$P \subseteq \Gamma(k), \qquad \mathrm{cl}(P) \setminus P = \varnothing, \tag{11}$$

unless the expansion is explicitly declared. This prevents a seemingly local edit from silently releasing unrelated parameters.

4.4 Scientific Validity Checks and Candidate Ranking

The pre-execution check rejects actions outside their stage and high-risk actions without bounds. It also rejects or requires explicit constraints for strongly correlated parameter combinations, such as simultaneous refinement of occupancy and displacement parameters or poorly identifiable combinations of zero shift, lattice parameters, and wavelength. After execution, $H(s_t, a_t, s'_{t+1}) = 0$ if a new physical violation appears, R_{wp} regresses by more than 2%, the unexplained-peak ratio increases by more than 0.02, the maximum absolute correlation exceeds 0.98, or any declared prediction fails.

Among states that pass all mandatory checks, the current implementation ranks candidates with

$$U(s) = -\Bigg[\frac{R_{\mathrm{wp}}(s)}{\max(R_{\mathrm{exp}}(s), \epsilon)} + 2R(s) + 3Q(s) + 0.002K(s) + 10\big[\,C_{\max}(s) - 0.90\,\big]_+ + 10|v(s)|\Bigg], \tag{12}$$

where R is residual score, Q unexplained-peak ratio, K parameter count, and $C_{\max}$ maximum absolute correlation. Here $\epsilon > 0$ is a numerical stabilizer. Weights are transparent policy constants rather than learned scientific truth. Equation 8 uses this score only after all mandatory checks pass.

4.5 Complete Refinement History

Each run record contains $(a_t, s_t, s'_{t+1}, \mathrm{dec}_t, O_{t+1})$, where dec_t is the acceptance decision and O_{t+1} lists the corresponding code, backend inputs, outputs, and evidence artifacts. The history records the proposed action, supporting observations, before/after states, validation results, and output-file paths for every accepted or rejected step. Retaining rejected attempts makes the final result reproducible and allows scientists to review which hypotheses were tried, why they failed, and how the accepted refinement path was obtained.

5 XRDBENCH

5.1 From Bounded Questions to Executable Workflows

An autonomous diffraction analysis combines local scientific judgments with implementation, backend use, evidence preservation, and termination. Endpoint-only evaluation reveals whether a workflow succeeded but obscures why; bounded questions isolate the failure but cannot establish autonomy. XRDBench therefore contains two complementary parts that evaluate the same process at different levels of integration:

$$\underbrace{\mathcal{B}_Q}_{\text{bounded XRD questions}} \longrightarrow \underbrace{\mathcal{B}_E}_{\text{state-changing workflows}}, \tag{13}$$

where the arrow denotes increasing task integration, not information flow.

The first part is XRDBench-QA. Here *QA* denotes question answering under bounded evidence and output requirements, not only multiple choice. Thirty cases require exact option sets, 40 require structured evidence-grounded reports, and 30 require quantitative outcomes plus reports. XRDBench-E2E contains 34 open workflows in which the agent must inspect files, implement and execute an analysis, revise failures, and package the evidence supporting its answer. Figure 2 summarizes the two tracks, their capability strata, and their task families; the complete task inventory is provided in the supplementary material.

The QA track uses three evaluation forms: exact matching for 30 scientific decisions, reference-guided assessment for 40 structured diagnoses, and recomputed objective metrics plus report assessment for 30 quantitative outcomes. E2E instead evaluates executed reports and artifacts through execution and physical gates, task-specific numerical metrics, and scientific judging. The complete protocol table is included in the supplementary material.

5.2 XRDBench-QA

The fixed 30/40/30 Easy/Medium/Hard split controls response complexity rather than defining a universal XRD taxonomy. The nine capabilities span decision control, evidence interpretation, and quantitative recovery; 70 cases are deterministic and 30 use experimental IUCr quantitative-phase and Dara phase-identification references [9, 22]. Public inputs contain pattern or trajectory evidence and a response schema; answers, numerical references, and rubrics remain evaluator-side.

For an objective metric m, baseline b, and oracle o, higher- and lower-is-better utilities are

$$u_{\uparrow}(m) = \mathrm{clip}\left(\frac{m-b}{o-b}, 0, 1\right), \quad u_{\downarrow}(m) = \mathrm{clip}\left(\frac{b-m}{b-o}, 0, 1\right). \tag{14}$$

Parameter recovery combines mechanism detection and normalized errors; QPA combines phase-set F1, fraction error/closure, amorphous accuracy, and report quality; phase identification combines phase-set F1, indexed coverage, unknown handling, and report quality. With normalized case scores $u_j^{(Q)} \in [0, 1]$, the QA track score is $S_Q = \sum_{j=1}^{100} u_j^{(Q)}$.

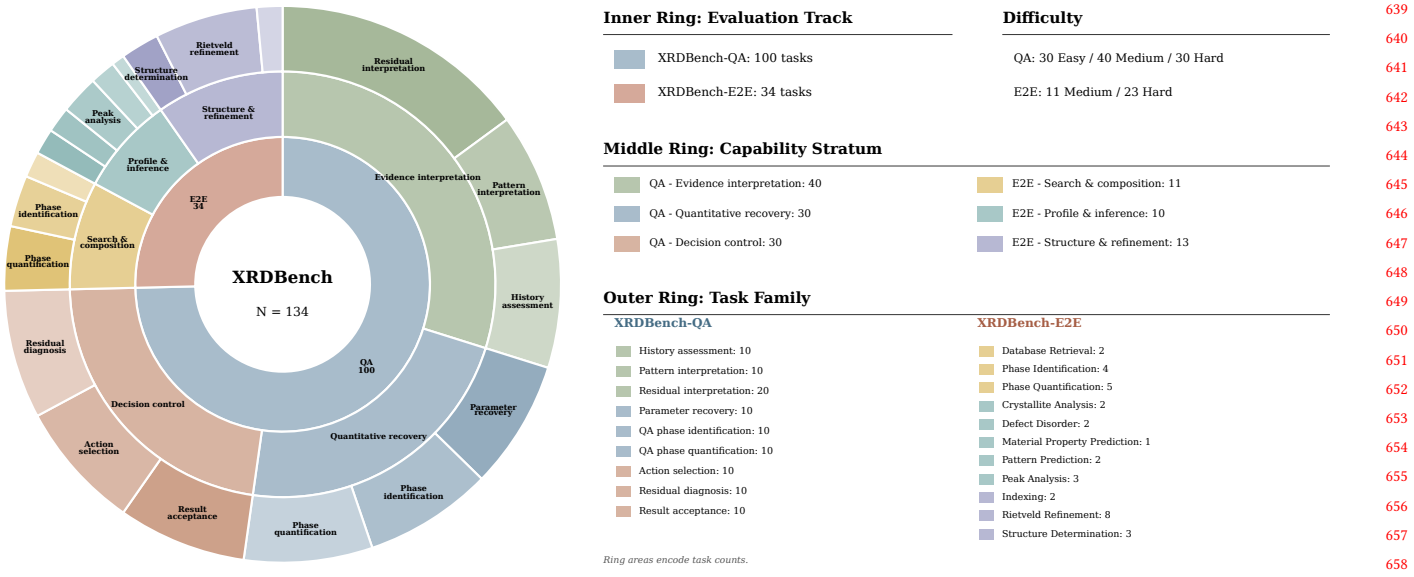


Figure 2: Composition of XRDBench. The inner ring separates the 100 QA tasks from the 34 executable E2E workflows; the middle ring groups related capabilities; and the outer ring shows nine QA capability categories and eleven E2E task families. Ring areas encode task counts, while difficulty is reported separately because it is not a task taxonomy.

5.3 XRDBench-E2E

The E2E track contains 34 unique measured patterns and source phase sets: 30 Cu Kα patterns from two opXRD sources [13] and four neutron examples distributed with FullProf. It includes 11 Medium and 23 Hard workflows. After license/provenance review, patterns and instrument metadata are normalized into public workspaces, while source labels and numerical references remain evaluator-side. The eleven scored families are retained separately; only the radar aggregates phase identification with retrieval and crystallite analysis with disorder, yielding nine reporting domains.

Each fresh, networkless workspace provides public inputs, native file/code tools, task-relevant XRD procedures, GSAS-II, FullProf, and numerical libraries. The agent must deliver `final_report.md` plus task-specific executed code and evidence. An *execution gate* checks input inspection, state change, successful execution, evidence, deliverables, and oracle isolation; an independent *physical gate* rejects numerical or crystallographic contradictions. Let $g_j = g_j^{\text{exec}} g_j^{\text{phys}} \in \{0,1\}$ be their joint outcome. Let $\phi_{A,j}, \phi_{M,j}, \phi_{J,j} \in [0,1]$ denote artifact integrity, task-specific numerical correctness, and evidence-grounded scientific judging, respectively. The case and track scores are

$$S_{E,j} = 100 g_j \left(0.15\phi_{A,j} + 0.55\phi_{M,j} + 0.30\phi_{J,j}\right), \qquad S_E = \frac{1}{34}\sum_{j=1}^{34} S_{E,j}. \tag{15}$$

5.4 Aggregation and Reproducibility

We primarily report (S_Q, S_E) and use

$$S_{\text{overall}} = 0.5 S_Q + 0.5 S_E \tag{16}$$

as the overall score so the 100:34 count ratio does not define scientific importance. All runs record trajectories, commands, artifacts, time, model turns, tool calls, and tokens; public cases enable audit, while future claims require sequestered extensions [28].

6 EXPERIMENTS

6.1 Experimental Setup

We evaluate GPT-5.6 Sol, Terra, and Luna; MiMo-V2.5-Pro and MiMo-V2.5; DeepSeek-V4-Pro and Flash; MiniMax-M2.7; Claude Haiku 4.5; and Gemini 3.1 Pro Low. Each model–task pair is executed once. On XRDBench-QA, each case permits at most 20 tool calls and 8,192 output tokens per turn; GPT-5.6 Luna evaluates rubric-based responses, while exact and objective scores are computed directly. On XRDBench-E2E, each workflow permits at most 50 agent steps and 32,768 output tokens per turn; GPT-5.6 Sol supplies the scientific-judgment component after execution and physical gates and objective metrics are evaluated. The complete protocol is provided in the supplementary material.

6.2 Research Questions and Statistical Analysis

We ask: **RQ1**, which XRD capabilities current models master; **RQ2**, whether they compose into reliable workflows; **RQ3**, how ranks and failure modes change with integration; and **RQ4**, how efficiency and component choices affect performance. The selected records contain 1,340 model–task runs, 22,135 model turns, and 625.6M selected-trajectory tokens: QA contributes 9,794 turns/198.2M tokens and E2E 12,341 turns/427.4M tokens. All intervals use 20,000 stratified task-bootstrap replicates (seed 20260824); model differences use paired resampling within track. Thus intervals describe task variation, not decoding stochasticity.

7 RESULTS

Figure 3 compares the two tracks and their aggregate without duplicating the complete numerical leaderboard, which is provided in the supplementary material.

7.1 Scientific Capability and Workflow Integration

The ten-model mean falls by 8.22 points, from 61.89 on XRDBench-QA to 53.67 on XRDBench-E2E, and seven of ten models score lower on E2E. Because the tracks contain different tasks, this is a descriptive integration gap rather than a paired difficulty estimate; together with completion and gate failures, it shows that executable XRD analysis remains substantially more challenging than bounded scientific questions. On QA, models are strongest at refinement-history assessment (88.0) and result acceptance (81.0), but weak at next-action selection (44.0) and quantitative phase analysis (33.0). All models miss high-risk occupancy/zero–cell coupling cases and the amorphous-content condition in the hardest QPA case; nine accept a lower-R_{wp} state that introduces an unexplained peak.

E2E exposes different bottlenecks. Indexing is weakest (37.72) despite a 90% gate rate: agents usually execute an indexing workflow but recover inaccurate cells or HKL assignments. Rietveld refinement averages 48.58 and has the lowest joint gate-passage rate (71.25%), indicating incomplete or invalid evidence packages; microstructure/disorder is also difficult (47.91). Figure 4 places the nine QA capabilities and nine E2E reporting domains on one profile per model.

7.2 Rank Transfer, Completion, and Efficiency

QA and E2E scores are moderately correlated (Pearson $r = 0.557$; Spearman $\rho = 0.648$), but rankings change materially. DeepSeek-V4-Pro rises from seventh on QA to third on E2E and Gemini from tenth to fifth, whereas MiMo-V2.5-Pro, MiMo-V2.5, and DeepSeek-V4-Flash lose 21.8–27.6 points. The overall estimates of Sol (81.09) and Terra (79.41) differ by 1.68 points with paired 95% interval $[-1.46, 4.80]$; neither is significantly ahead.

Figure 5 pools resource use over all 134 tasks per model and removes frontier lines: duration, comparable model turns, selected-trajectory tokens, and cost are reported directly against the overall score.

7.3 Failure Analysis

Trace inspection exposes three recurring E2E failure types. First, *scientific outcome failures* occur after genuine execution: indexing often produces inaccurate cells or HKL assignments despite passing the gate, while refinement may improve fit statistics without a valid structural result. Second, *execution and evidence failures* prevent scoring or completion: 46/340 runs fail at least one of the execution or physical gates, and 117/340 fail at least one full-completion requirement, commonly because code, backend outputs, or required evidence is missing or invalid. Third, *termination failures* consume almost the full 50-step budget without producing a valid deliverable. These traces explain why aggregate QA competence transfers only partially to autonomous workflows and separate runnable-but-wrong science from incomplete execution.

7.4 Component Effects

All six tested components improve GPT-5.6 Luna on their targeted QA settings (Figure 6). Result-acceptance checks and final scientific review yield the largest gains; the smaller targeted sets make these diagnostic rather than E2E-wide causal estimates.

8 DISCUSSION

QA and E2E expose complementary failure modes. The former diagnoses whether a model recognizes a valid action or scientific outcome; the latter adds implementation, backend operation, evidence preservation, physical validation, and delivery. Low Rietveld joint gate passage indicates incomplete or physically invalid workflows, whereas low Indexing accuracy despite high joint gate passage indicates incorrect science after genuine execution. The former calls for stronger execution contracts and stopping controls; the latter for better candidate generation and uncertainty-aware algorithms.

Deterministic controls remain necessary even for frontier models: QA cases show models overriding a hard residual regression when a familiar scalar improves, while E2E shows fluent reports without reproducible evidence. AutoXRD therefore treats backend outputs, artifacts, and physical checks as first-class evidence. Deployment should report track scores, gate rate, runtime, tokens, and backend reliability together rather than a single leaderboard rank.

9 LIMITATIONS

Several E2E domains contain only one to three cases, so domain profiles are descriptive; the 34 inputs are unique measured patterns/source phase sets, not independently audited as chemically unique materials. One solver pass per task measures task variation rather than decoding stochasticity. Report quality is judged once by GPT-5.6 Sol, an evaluated solver; objective metrics and gates reduce but do not remove evaluator dependence, and expert agreement remains future work. The 30 X-ray/four-neutron composition motivates the broader powder-diffraction claim. The public suite permits future contamination, and selected-trajectory accounting is not total provider spend. Sequestered extensions, broader instruments/materials, repeated seeds, and blinded crystallographer review are needed before safety-critical deployment.

10 CONCLUSION

AutoXRD combines LLM planning with crystallographic backends, evidence-preserving execution, and deterministic checks; XRD-Bench evaluates the same process from diagnostic question answering to end-to-end workflows. Across ten models and 1,340 runs, QA competence only moderately predicts autonomous performance. Indexing is often executed but inaccurate, whereas Rietveld refinement is limited chiefly by workflow completion and validity. Reliable diffraction agents must expose the executable evidence and invariants licensing each conclusion, not merely a plausible final report.

ETHICAL CONSIDERATIONS

Incorrect autonomous diffraction conclusions can affect materials decisions and downstream experiments. AutoXRD is decision support rather than expert certification: it exposes uncertainty and

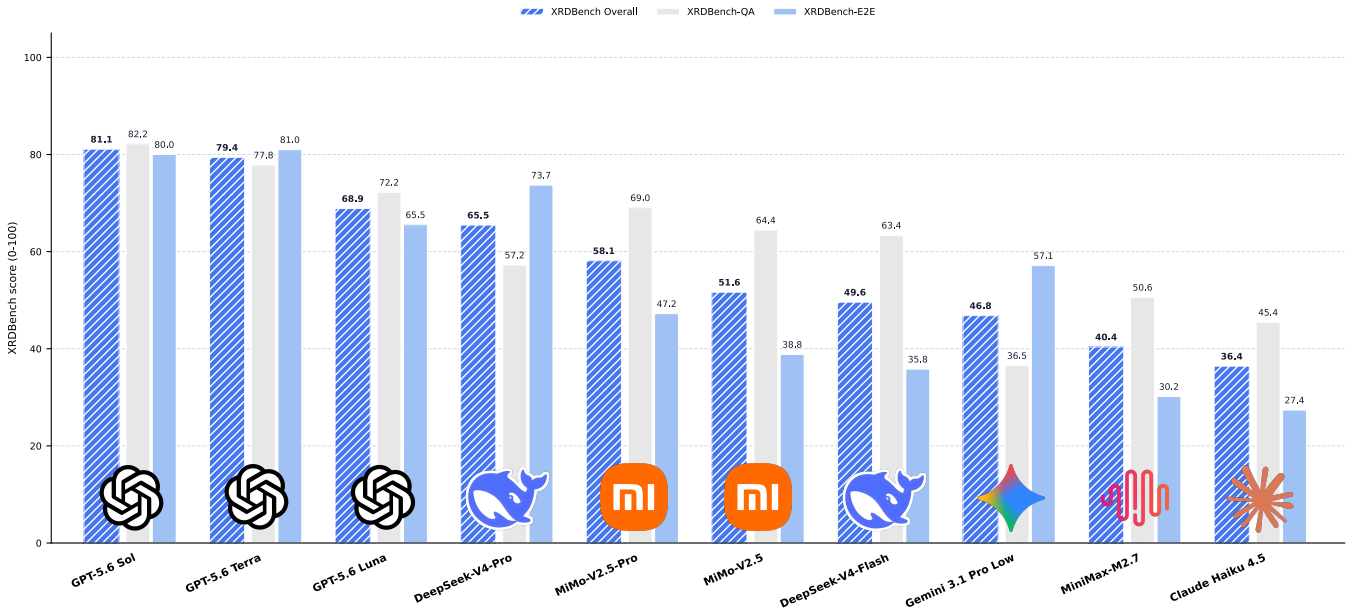


Figure 3: Overall performance, ordered by the overall score. The wider, hatched bar denotes Overall, followed by XRDBench-QA and XRDBench-E2E.

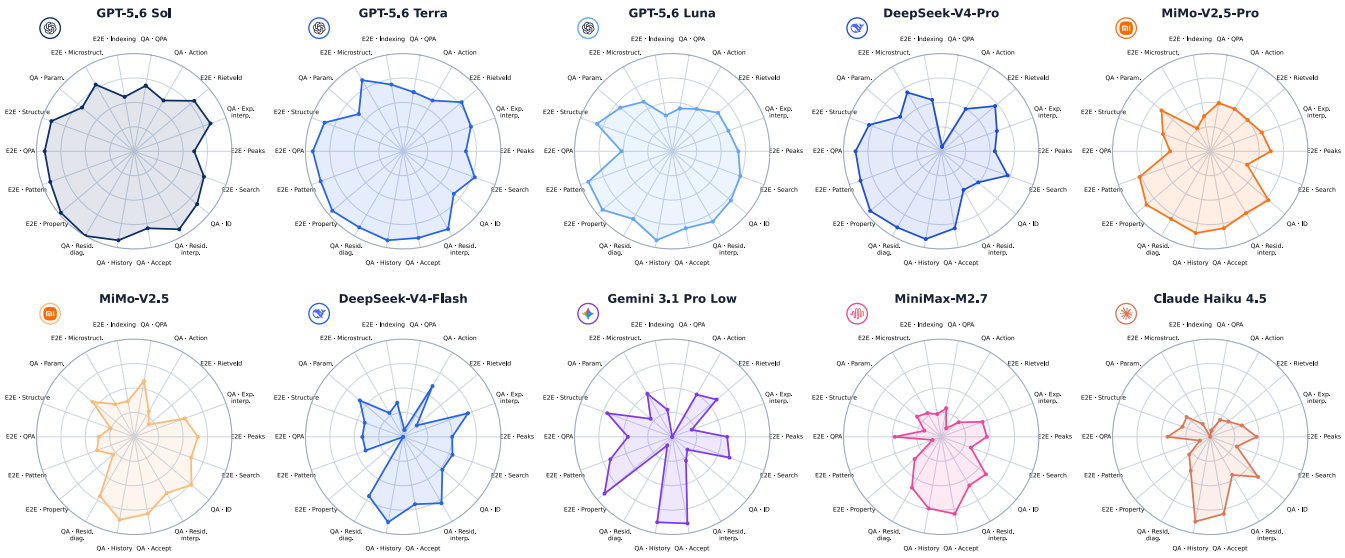


Figure 4: Fine-grained performance across 18 dimensions: nine QA capabilities and nine E2E reporting domains. Each model has a separate 0–100 radar; numeric radial labels are omitted. Figure 2 defines the underlying task families, and the supplementary material provides the complete mapping.

validation failures, while users remain responsible for provenance, safety, calibration, licensing, and independent scientific review.

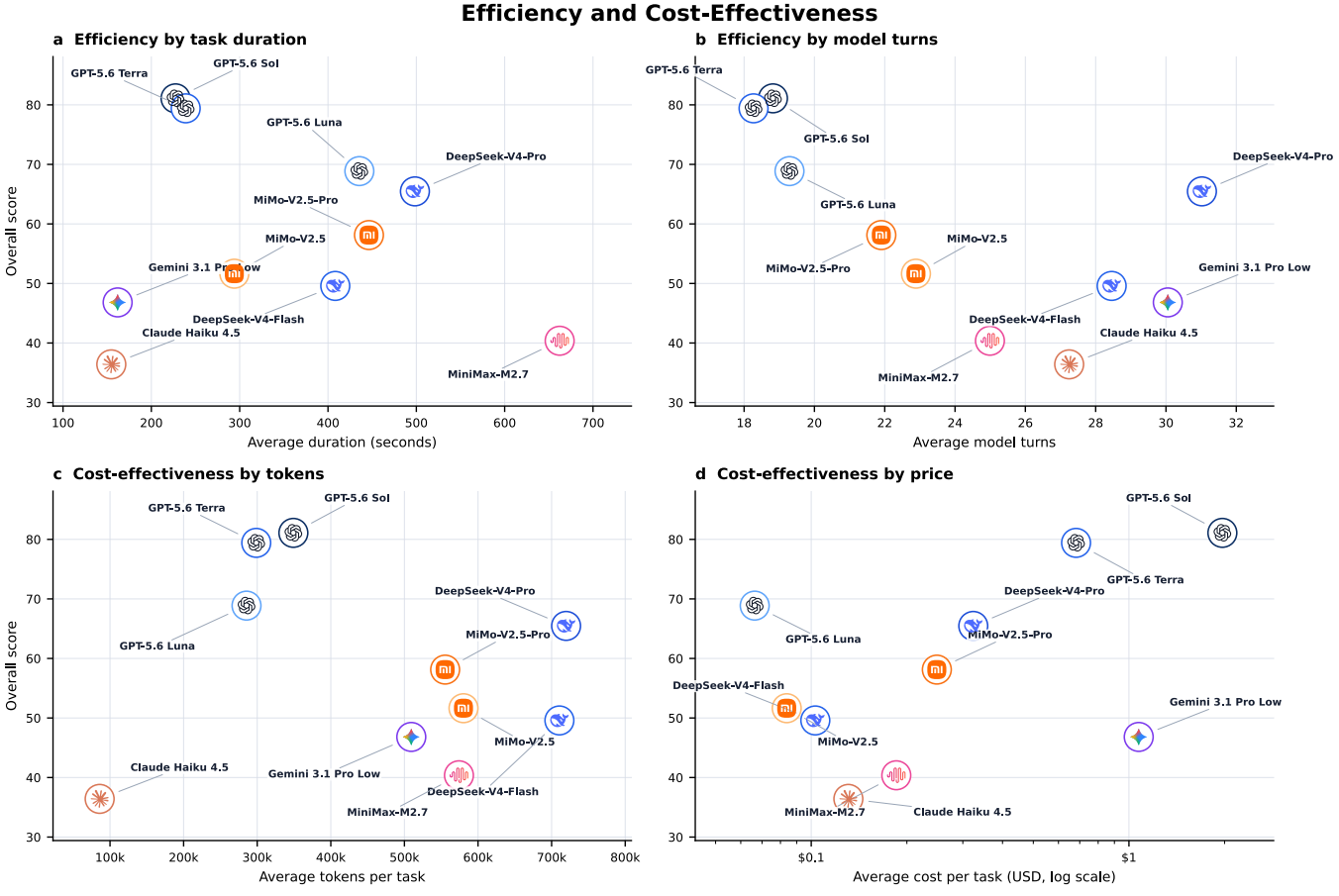


Figure 5: Full-benchmark efficiency and cost-effectiveness. Resource means pool all 134 task runs per model; the vertical axis is the overall score. Prices cover selected trajectories rather than superseded attempts.

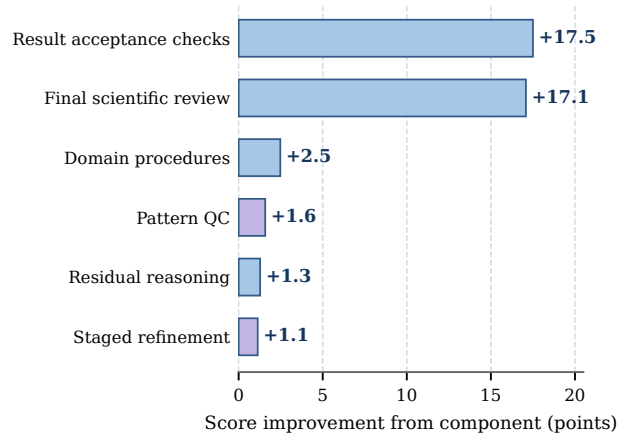


Figure 6: GPT-5.6 Luna component ablations on targeted XRDBench-QA cases. Bars report the full-system score minus the score after removing the named component; Pattern QC and staged refinement use prospective complementary holdouts. Full control/removal scores and case counts are in the supplementary material.

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